

Line Buffer

A memory buffer used to hold one line of video. If the horizontal resolution of the screen is 640 pixels, and the colour space is RGB, the line buffer would have to be 640 locations long by 3 bytes wide. This amounts to one location for each pixel and each colour plane. Line buffers are typically used in filtering algorithms.

MIP Mapping

‘Multum in Parvum’ means ‘many in one’. A method of increasing the quality of a texture map by applying different-resolution texture maps for different objects in the same image, depending on their size and depth. If a texture-mapped polygon is smaller than the texture image itself, the texture map will be undersampled during rasterisation. As a result, the texture mapping will be noisy and ‘sparkly’. MIP mapping removes this effect.

Rendering

The process of creating life-like images on a screen using mathematical models and formulas to add shading, colour and lamination to a 2D or 3D wireframe.

Rendering Engine

Generically applies to the part of the graphics engine that draws 3D primitives, usually triangles or other simple polygons. In most implementations, the rendering engine is responsible for interpolation of edges and ‘filling in’ the triangle.

Texture Filtering

Removing the undesirable distortion of a raster image, also called aliasing artefacts, such as sparkles and blockiness, through interpolation of stored texture images.

Texture Mapping

Texture mapping is based on a stored bitmap consisting of texture pixels, or texels. It consists of wrapping a texture image onto an object to create a realistic representation of the object in 3D space. The object is represented by a set of polygons, usually triangles.